LEARNER GUIDE

Solving linear inequalities

Inequalities · Lesson 2. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

You can solve an inequality in the same way as an equation: do the **same to both sides** until x stands alone. Adding, subtracting, and multiplying or dividing by a **positive** number keep the inequality true, with the symbol facing the same way. There is one difference. Multiplying or dividing both sides by a **negative** number reflects both sides across 0 on the number line, which reverses their order, so the symbol must **reverse** too. You can always **check** an answer by testing a value.

What you are learning to notice

- Most steps keep the symbol the same: adding, subtracting, and multiplying or dividing by a positive number.
- Multiplying or dividing by a **negative** number reverses the symbol: < becomes >, and ≤ becomes ≥.
- Why: 2 < 5, but –2 > –5. Multiplying by –1 reflects both numbers across 0, so their order reverses.
- Check with a test value: a value inside your answer makes the original inequality true; a value outside makes it false.

Moving through it, one step at a time

Write each step on a new line, as you would for an equation. Undo the added or subtracted number first, then the number multiplying x. Before you divide, look at the sign of that number: if it is negative, reverse the symbol. When x stands alone, show the answer on a number line, then test one value inside it and one outside it in the original inequality.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Two steps, no reversal

Solve $3x + 2 > 11$.

1. Subtract 2 from both sides.	$3x > 9$
2. Divide both sides by 3. It is positive, so the symbol stays.	$x > 3$
3. Check with $x = 4$: $3 \times 4 + 2 = 14$, and $14 > 11$.	true ✓

$x > 3$. Open circle at 3, arrow to the right.

Example 2 — When the symbol reverses

Solve $5 - 2x \geq 1$.

1. Subtract 5 from both sides.	$-2x \geq -4$
2. Divide both sides by –2. It is negative, so reverse ≥ to ≤.	$x \leq 2$
3. Check with $x = 0$: $5 - 0 = 5$, and $5 \geq 1$.	true ✓

$x \leq 2$. Filled circle at 2, arrow to the left.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (solving one- and two-step equations, and the inequality symbols from Lesson 1), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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